

1. ORIENTED LATTICE ANIMAL ESTIMATES

For $a \leq b$, write

$$\llbracket a, b \rrbracket := [a, b] \cap \mathbb{Z}.$$

For integers $N, L \geq 0$, set

$$\mathcal{X}_L := \llbracket -L, L \rrbracket^2, \quad \mathbb{S}_{N,L} := \llbracket 0, N \rrbracket \times \mathcal{X}_L.$$

For $\ell \in \llbracket 0, N \rrbracket$, set

$$T_\ell := T_\ell^{N,L} := \{\ell\} \times \mathcal{X}_L.$$

Define

$$\vec{E} := \vec{E}_{N,L} := \{ \langle (m, x), (m+1, y) \rangle : m \in \llbracket 0, N-1 \rrbracket, x, y \in \mathcal{X}_L, |x - y|_\infty \leq 1 \}$$

and

$$E := E_{N,L} := \{ \{u, v\} : u, v \in \mathbb{S}_{N,L}, u \neq v, |u - v|_\infty \leq 1 \}.$$

Fix integers $N, L \geq 3$ with $N \leq e^L$. Write vertices as $v = (m, x)$ with $m \in \llbracket 0, N \rrbracket$ and $x \in \mathcal{X}_L$.

Let $\omega = (\omega_e)_{e \in \vec{E}} \in \{-1, 1\}^{\vec{E}}$. We say that an edge e is ω -open if $\omega_e = 1$.

A non-oriented path is a finite sequence $\gamma = (v_0, \dots, v_r)$ of vertices in $\mathbb{S}_{N,L}$ such that $\{v_i, v_{i+1}\} \in E$ for every $0 \leq i < r$. An oriented open path is a finite sequence $\gamma = (v_0, \dots, v_r)$ such that $\langle v_i, v_{i+1} \rangle \in \vec{E}$ and $\omega_{\langle v_i, v_{i+1} \rangle} = 1$ for every $0 \leq i < r$. In both definitions, the length 0 path is allowed.

For vertices u, v , write $u \leftrightarrow v$ if there exists an oriented open path from u to v . Write $A \leftrightarrow B$ for sets A, B if there exist $a \in A$ and $b \in B$ such that $a \leftrightarrow b$.

Define the event

$$\mathcal{C} := \{T_0 \leftrightarrow T_N\}.$$

Define the reached set

$$B := B(\omega) := \{v \in \mathbb{S}_{N,L} : T_0 \leftrightarrow v\}.$$

Since the length 0 path is allowed, $T_0 \subset B(\omega)$ for every ω .

Given $B = B(\omega)$, define

$$F := F(\omega) := \left\{ v \in \mathbb{S}_{N,L} \setminus B : \exists \text{ a non-oriented path } \gamma \text{ from } v \text{ to } T_N \right. \\ \left. \text{in } \mathbb{S}_{N,L} \text{ such that the vertices of } \gamma \text{ avoid } B \right\}.$$

Define the oriented inner edge-boundary

$$\vec{\partial}_E^{\text{in}} B := \vec{\partial}_E^{\text{in}} B(\omega) := \{ \langle u, v \rangle \in \vec{E} : u \in B(\omega), v \in F(\omega) \},$$

and the corresponding inner boundary tails

$$\vec{\partial}^{\text{in}} B := \vec{\partial}^{\text{in}} B(\omega) := \{ u \in B(\omega) : \exists v \in F(\omega) \text{ with } \langle u, v \rangle \in \vec{E} \}.$$

Equivalently, $u \in \vec{\partial}^{\text{in}} B(\omega)$ if and only if $u \in B(\omega)$ and there exists an oriented edge $\langle u, v \rangle \in \vec{E}$ with $v \in F(\omega)$; that is, v admits a non-oriented path to T_N inside $\mathbb{S}_{N,L}$ whose vertex set is disjoint from $B(\omega)$.

Define the enlarged inner boundary

$$\partial^{\text{in}} B(\omega) := \{ u \in B(\omega) : \exists v \in F(\omega) \text{ with } \{u, v\} \in E \}.$$

Since every oriented edge has its underlying non-oriented edge in E , we have $\vec{\partial}^{\text{in}} B \subset \partial^{\text{in}} B$.

Lemma 1.1. *On $\mathcal{C}^c$, every oriented path in $(\mathbb{S}_{N,L}, \vec{E})$ from T_0 to T_N contains at least one edge of $\vec{\partial}_E^{\text{in}} B$.*

Proof. Let $\pi = (v_0, v_1, \dots, v_r)$ be an oriented path in $(\mathbb{S}_{N,L}, \vec{E})$ with $v_0 \in T_0$ and $v_r \in T_N$. Since every oriented edge increases the time coordinate by one, necessarily $r = N$. In particular, $r \geq 1$ and no vertex is repeated. Since $\mathcal{C}^c$ holds, we have $T_N \cap B = \emptyset$, and in particular $v_r \notin B$. Moreover, $v_r \in F$ because the length 0 path (v_r) is a non-oriented path from v_r to T_N whose vertex set is disjoint from B .

Let i be the smallest index such that $v_i \in F$. Then $i \geq 1$ because $v_0 \in T_0 \subset B$ and $F \subset \mathbb{S}_{N,L} \setminus B$. If $v_{i-1} \notin B$, then concatenating the underlying non-oriented edge $\{v_{i-1}, v_i\}$ with a non-oriented path from v_i to T_N avoiding B (which exists since $v_i \in F$) yields a non-oriented path from v_{i-1} to T_N avoiding B , hence $v_{i-1} \in F$, a contradiction. Thus $v_{i-1} \in B$ and $v_i \in F$, and therefore $\langle v_{i-1}, v_i \rangle \in \vec{\partial}_E^{\text{in}} B$. $\square$

Lemma 1.2. *Assume $\omega \in \mathcal{C}^c$. Every vertical fiber $\llbracket 0, N \rrbracket \times \{x\}$, $x \in \mathcal{X}_L$, meets $\vec{\partial}^{\text{in}} B$. In particular,*

$$|\vec{\partial}^{\text{in}} B(\omega)| \geq |\mathcal{X}_L| = (2L+1)^2.$$

Moreover, if $x \in \mathcal{X}_L$, $m \in \llbracket 1, N \rrbracket$, and $(m, x) \in \partial^{\text{in}} B \setminus \vec{\partial}^{\text{in}} B$, then either $(m-1, x) \in \vec{\partial}^{\text{in}} B \cup F$, or $m \geq 2$ and $(m-2, x) \in \vec{\partial}^{\text{in}} B \cup F$.

Proof. Fix $x \in \mathcal{X}_L$ and consider the vertical oriented path

$$\pi_x := ((0, x), (1, x), \dots, (N, x)).$$

It is an oriented path in $(\mathbb{S}_{N,L}, \vec{E})$ from T_0 to T_N . By Lemma 1.1, π_x contains an edge in $\vec{\partial}_E^{\text{in}} B$. Since the only edges of π_x are $\langle (m, x), (m+1, x) \rangle$, there exists $m_x \in \llbracket 0, N-1 \rrbracket$ such that

$$\langle (m_x, x), (m_x + 1, x) \rangle \in \vec{\partial}_E^{\text{in}} B.$$

In particular, $(m_x, x) \in \vec{\partial}^{\text{in}} B$. The map $x \mapsto (m_x, x)$ is injective, hence

$$|\vec{\partial}^{\text{in}} B| \geq |\mathcal{X}_L| = (2L+1)^2.$$

Now let $x \in \mathcal{X}_L$ and $m \in \llbracket 1, N \rrbracket$ satisfy $(m, x) \in \partial^{\text{in}} B \setminus \vec{\partial}^{\text{in}} B$. Choose $v = (r, y) \in F$ adjacent to (m, x) in the non-oriented graph $(\mathbb{S}_{N,L}, E)$. Since $(m, x) \notin \vec{\partial}^{\text{in}} B$, the time coordinate of v cannot be $m+1$, for otherwise $\langle (m, x), v \rangle \in \vec{E}$ and (m, x) would belong to $\vec{\partial}^{\text{in}} B$. Hence $r \in \{m, m-1\}$.

If $r = m$, then $\langle (m-1, x), v \rangle \in \vec{E}$. If $(m-1, x) \in B$, this oriented edge from $(m-1, x)$ to F shows that $(m-1, x) \in \vec{\partial}^{\text{in}} B$. If instead $(m-1, x) \notin B$, concatenating the underlying non-oriented edge $\{(m-1, x), v\}$ with a non-oriented path from v to T_N avoiding B shows that $(m-1, x) \in F$.

If $r = m-1$, then $m \geq 2$, since otherwise $v \in T_0 \subset B$, contradicting $v \in F$. In this case $\langle (m-2, x), v \rangle \in \vec{E}$. The same argument gives either $(m-2, x) \in \vec{\partial}^{\text{in}} B$ or $(m-2, x) \in F$. $\square$

1.1. From oriented lattice animal to lattice animal estimates.

Theorem 1.3. *Assume $\omega \in \mathcal{C}^c$. It holds*

$$|\partial^{\text{in}} B| \leq 3|\vec{\partial}^{\text{in}} B|.$$

We are now in a position to prove Theorem 1.3.

Proof of Theorem 1.3. Fix $x \in \mathcal{X}_L$ and write

$$I_x := \{m \in \llbracket 0, N \rrbracket : (m, x) \in \partial^{\text{in}} B\}, \quad J_x := \{m \in \llbracket 0, N \rrbracket : (m, x) \in \vec{\partial}^{\text{in}} B\},$$

and

$$G_x := \{m \in \llbracket 0, N \rrbracket : (m, x) \in F\}.$$

Clearly $J_x \subset I_x$ and $I_x \cap G_x = \emptyset$. We regard G_x as empty outside $\llbracket 0, N \rrbracket$ when writing expressions such as $q+1 \in G_x$. We prove that $|I_x| \leq 3|J_x|$.

We first note a simple consequence of the definition of F . If $t \in G_x$ and $t \geq 1$, then either $t-1 \in G_x$ or $t-1 \in J_x$. Indeed, if $(t-1, x) \in B$, then the vertical edge $\langle (t-1, x), (t, x) \rangle \in \vec{E}$

shows that $t - 1 \in J_x$. If $(t - 1, x) \notin B$, concatenating the underlying non-oriented edge $\{(t - 1, x), (t, x)\}$ with a non-oriented path from (t, x) to T_N avoiding B shows that $t - 1 \in G_x$. Since $T_0 \subset B$, every nonempty connected component $\llbracket a, b \rrbracket$ of G_x has $a \geq 1$ and $a - 1 \in J_x$.

For each $q \in J_x$, define $h(q)$ as follows. If $q + 1 \notin G_x$, set $h(q) = q$. If $q + 1 \in G_x$, let $h(q)$ be the upper endpoint of the connected component of G_x that contains $q + 1$. Set

$$K_q := \{q, h(q) + 1, h(q) + 2\} \cap \llbracket 0, N \rrbracket.$$

We claim that

$$I_x \subset \bigcup_{q \in J_x} K_q.$$

Let $m \in I_x$. If $m \in J_x$, then $m \in K_m$. Hence assume $m \notin J_x$. The case $m = 0$ is impossible, because $T_0 \subset B$ rules out an F -neighbor of $(0, x)$ at time 0, and any F -neighbor at time 1 would put 0 in J_x . Thus $m \geq 1$.

By Lemma 1.2, there exists $s \in \{m - 1, m - 2\} \cap (J_x \cup G_x)$, with the convention that $m - 2$ is available only when $m \geq 2$. Choose such an s . If $s \in J_x$, then $m = s + 1$ or $m = s + 2$. If $s + 1 \notin G_x$, then $h(s) = s$ and $m \in K_s$. If $s + 1 \in G_x$, the fact that $m \in I_x$ and $I_x \cap G_x = \emptyset$ forces the component of G_x starting at $s + 1$ to end before m ; hence $m = h(s) + 1$ or $m = h(s) + 2$, and again $m \in K_s$.

It remains to consider the case where $s \in G_x$. Let $\llbracket a, b \rrbracket$ be the connected component of G_x containing s . The observation above gives $a - 1 \in J_x$. Since $m = s + 1$ or $m = s + 2$ and $m \notin G_x$, we must have $m = b + 1$ or $m = b + 2$. Therefore $m \in K_{a-1}$.

Since each set K_q has at most three elements,

$$|I_x| \leq 3|J_x|$$

for every $x \in \mathcal{X}_L$. Since

$$|\partial^{\text{in}} B| = \sum_{x \in \mathcal{X}_L} |I_x|, \quad |\vec{\partial}^{\text{in}} B| = \sum_{x \in \mathcal{X}_L} |J_x|,$$

summing over $x \in \mathcal{X}_L$ yields

$$|\partial^{\text{in}} B| \leq 3|\vec{\partial}^{\text{in}} B|.$$

□

Recall that

$$\mathbb{S}_{N,L} := \llbracket 0, N \rrbracket \times \llbracket -L, L \rrbracket^2, \quad T_\ell := \{\ell\} \times \llbracket -L, L \rrbracket^2.$$

For the counting step we use $*$ -adjacency on $\mathbb{Z}^3$, i.e. for distinct vertices u, v we write $u \sim_* v$ when $|u - v|_\infty \leq 1$.

Lemma 1.4 (Finite-box Peierls lemma for $*$). *Let $U \subset \mathbb{S}_{N,L}$ be nonempty and $*$ -connected with $T_0 \subset U$ and $T_N \cap U = \emptyset$. Let H be the $*$ -connected component of $\mathbb{S}_{N,L} \setminus U$ which touches T_N , and assume $H \neq \emptyset$. Then*

$$V := \{u \in U : \exists h \in H \text{ with } u \sim_* h\}$$

is $$ -connected.*

Proof. Let $C := \mathbb{S}_{N,L} \setminus H$. We claim that C is $*$ -connected. Indeed, $U \subset C$, and U is $*$ -connected by assumption. If K is any $*$ -connected component of $\mathbb{S}_{N,L} \setminus U$ different from H , then K has a $*$ -neighbor in U . Otherwise, since $\mathbb{S}_{N,L}$ is $*$ -connected, a $*$ -path from K to $\mathbb{S}_{N,L} \setminus K$ would have a first edge leaving K , and its endpoint outside K would lie in $\mathbb{S}_{N,L} \setminus U$, contradicting maximality of the component K . Hence every such K is attached to U , so C is $*$ -connected.

Define the $*$ -boundary of H inside C by $\partial_* H := \{c \in C : \exists h \in H \text{ with } c \sim_* h\}$. The inclusion $V \subset \partial_* H$ follows from $U \subset C$. Conversely, if $c \in \partial_* H$ and $c \notin U$, then $c \in \mathbb{S}_{N,L} \setminus U$. Since $c \sim_* h$ for some $h \in H$, the vertex c lies in the same $*$ -connected component of $\mathbb{S}_{N,L} \setminus U$ as h , so $c \in H$, contradicting $c \in C$. Thus every vertex of $\partial_* H$ lies in U . Therefore, we have $\partial_* H = V$.

It remains to show that $\partial_* H$ is $*$ -connected. We use the following finite-box corollary of Timar's boundary-connectivity theorem; see Timar [1, Theorem 3]. If Q is a finite induced box in $\mathbb{Z}^3$ and $A \subset Q$ is nonempty and $*$ -connected, then the $*$ -outer boundary of A in any $*$ -connected component of $Q \setminus A$ is $*$ -connected. Equivalently, if both A and $Q \setminus A$ are nonempty and $*$ -connected, then

$$\partial_* A := \{z \in Q \setminus A : \exists a \in A \text{ with } z \sim_* a\}$$

is $*$ -connected.

We apply this corollary with $Q = \mathbb{S}_{N,L}$ and $A = H$. The set H is $*$ -connected by definition, and we proved above that $\mathbb{S}_{N,L} \setminus H = C$ is $*$ -connected. Both sets are nonempty. Hence $\partial_* H$ is $*$ -connected. Since $\partial_* H = V$, the required set V is $*$ -connected. $\square$

Lemma 1.5 (Peierls connectivity). *Assume $\omega \in \mathcal{C}^c$. The set $\partial^{\text{in}} B(\omega)$ is $*$ -connected.*

Proof. The set $B(\omega)$ is $*$ -connected: T_0 is $*$ -connected, and every $v \in B(\omega)$ is joined to T_0 by an oriented open path, whose underlying non-oriented path is a $*$ -path.

Also, $F(\omega)$ is exactly the union of the $*$ -connected components of $\mathbb{S}_{N,L} \setminus B(\omega)$ which touch T_N . Indeed, a vertex $v \notin B(\omega)$ belongs to $F(\omega)$ precisely when there is an $*$ -path from v to T_N avoiding $B(\omega)$. On $\mathcal{C}^c$ we have $T_N \subset F(\omega)$, so $F(\omega)$ is nonempty.

Applying Lemma 1.4 with $U = B(\omega)$ and $H = F(\omega)$ gives

$$\{u \in B(\omega) : \exists v \in F(\omega) \text{ with } u \sim_* v\}$$

is $*$ -connected. This set is $\partial^{\text{in}} B(\omega)$ by definition, since E is the restriction of $*$ -adjacency to $\mathbb{S}_{N,L}$. $\square$

1.2. Main counting claim.

Theorem 1.6. *There exists a constant $C < \infty$ such that, for all integers $N, L \geq 3$ with $N \leq e^L$ and every integer $k \geq 1$,*

$$\left| \left\{ \vec{\partial}^{\text{in}} B(\omega) : \omega \in \{-1, 1\}^{\vec{E}}, |\vec{\partial}^{\text{in}} B(\omega)| = k, \mathcal{C}(\omega)^c \right\} \right| \leq e^{Ck}.$$

Proof. Fix an integer $k \geq 1$. Let

$$\mathcal{S}_{N,L,k} := \left\{ \vec{\partial}^{\text{in}} B(\omega) : \omega \in \{-1, 1\}^{\vec{E}}, \mathcal{C}(\omega)^c, |\vec{\partial}^{\text{in}} B(\omega)| = k \right\}.$$

If $k < (2L+1)^2$, then Lemma 1.2 gives $\mathcal{S}_{N,L,k} = \emptyset$. We may therefore assume $k \geq (2L+1)^2$. We use the standard lattice-animal estimate: there exists $C_1 < \infty$, depending only on the dimension, such that for every integer $m \geq 1$,

$$|\{A \subset \mathbb{Z}^3 : |A| = m, A \text{ is } * \text{-connected, and } 0 \in A\}| \leq \exp(C_1 m).$$

For each $S \in \mathcal{S}_{N,L,k}$, let $a(S) \in S$ be its lexicographic minimum. The order is taken in the coordinates (m, x_1, x_2) . Set

$$S_0 := S - a(S) \subset \mathbb{Z}^3.$$

Then $0 \in S_0$ and $|S_0| = k$.

Fix $S \in \mathcal{S}_{N,L,k}$ and choose ω such that $S = \vec{\partial}^{\text{in}} B(\omega)$. On $\mathcal{C}^c$, Theorem 1.3 gives

$$|\partial^{\text{in}} B(\omega)| \leq 3|\vec{\partial}^{\text{in}} B(\omega)| = 3k.$$

Since $S \subset \partial^{\text{in}} B(\omega)$, the translated set

$$A := \partial^{\text{in}} B(\omega) - a(S)$$

contains S_0 . Thus

$$0 \in S_0 \subset A, \quad |A| \leq 3k.$$

By Lemma 1.5, the set $\partial^{\text{in}} B(\omega)$ is $*$ -connected. Consequently A is a $*$ -connected lattice animal containing the origin.

We first count the possible translated sets S_0 . If $|A| = m$, then $k \leq m \leq 3k$. There are at most $\exp(C_1 m)$ choices for such an A . After A is fixed, there are at most

$$\binom{m-1}{k-1} \leq 2^m$$

choices for a subset $S_0 \subset A$ with $0 \in S_0$ and $|S_0| = k$. Therefore

$$|\{S_0 : S \in \mathcal{S}_{N,L,k}\}| \leq \sum_{m=k}^{3k} \exp(C_1 m) 2^m \leq (3k) \exp((C_1 + \log 2) 3k).$$

Since $3k \leq \exp(2k)$ for $k \geq 1$,

$$|\{S_0 : S \in \mathcal{S}_{N,L,k}\}| \leq \exp((3(C_1 + \log 2) + 2)k).$$

It remains to count the possible anchors. The map $S \mapsto (a(S), S_0)$ is injective, and $a(S) \in \mathbb{S}_{N,L}$. Hence

$$|\mathcal{S}_{N,L,k}| \leq |\mathbb{S}_{N,L}| \cdot |\{S_0 : S \in \mathcal{S}_{N,L,k}\}|.$$

We have $|\mathbb{S}_{N,L}| = (N+1)(2L+1)^2$ and $N \leq e^L$. Since $L \geq 2$ and $k \geq (2L+1)^2$,

$$\log |\mathbb{S}_{N,L}| = \log(N+1) + 2 \log(2L+1) \leq (L+1) + 2(2L+1) \leq (2L+1)^2 \leq k,$$

and therefore $|\mathbb{S}_{N,L}| \leq \exp(k)$.

Combining the bounds,

$$|\mathcal{S}_{N,L,k}| \leq \exp(k) \cdot \exp((3(C_1 + \log 2) + 2)k) = \exp((3(C_1 + \log 2 + 1))k).$$

□

REFERENCES

- [1] A. Timar, *Boundary-connectivity via graph theory*, Proc. Amer. Math. Soc. 141 (2013), no. 2, 475–480.